Algebra II with Trigonometry Chapter 7.1

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4564659_alg-2trig-h-chp-7-1-note.pdf

Problems

1. Simplify mentally:

$$\frac{1-\sin x}{\cos x}+\frac{\cos x}{1-\sin x}.$$

2. If

$$\tan\theta+\sec\theta=7,$$

find the exact value of

$$\tan\theta-\sec\theta.$$

3. Evaluate without a calculator:

$$\frac{\sin 10^\circ+\sin 50^\circ+\sin 70^\circ}{\cos 10^\circ+\cos 50^\circ+\cos 70^\circ}.$$

4. For acute θ, suppose

$$\sin\theta+\cos\theta=\frac{5}{4}.$$

Find the exact value of

$$\tan\theta+\cot\theta.$$

5. Simplify:

$$\frac{1+\tan x+\sec x}{1+\tan x-\sec x}.$$

6. Let

$$E=\frac{\sin^4 x-\cos^4 x}{\sin^2 x-\cos^2 x}+\frac{1}{\sec^2 x}+\frac{1}{\csc^2 x}.$$

Determine the constant value of E.